Attention Based Spectral Profile Representation for Hyperspectral Image Classification

Shashi Kiran Chandrappa
Dept. of Analytics
Fairfield University, CT, USA
schandrappa@student.fairfield.edu

Sidike Paheding
Dept. of Computer Science and Engineering
Fairfield University, CT, USA
spaheding@fairfield.edu

Abel A. Reyes-Angulo
Dept. of Applied Computing
Michigan Technological University, MI, USA
areyesan@mtu.edu

Almabrok Essa
Dept. of Math, Computer Sci., and Data Sci.
John Carroll University, OH, USA
aessa@jcu.edu

NAECON 2025 - IEEE National Aerospace and Electronics Conference | 979-8-3315-3813-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/NAECON65708.2025.11235401

***Abstract*—We introduce an Attention-based Spectral Profile Representation (ASPR) leveraging multi-head attention mechanisms for pixel-wise hyperspectral image classification. By incorporating attention-driven importance scores, ASPR effectively identifies and preserves essential spectral regions, discarding redundant bands. By leveraging attention scores of spectral signatures, the proposed method effectively captures key spectral features, reduces redundancy, and improves classification accuracy. Experimental findings indicate that ASPR offers superior performance compared to conventional and current deep learning approaches for classifying hyperspectral images.**

***Index Terms*—Hyperspectral Image Classification, deep learning, self-attention, vision transformer**

I. Introduction

Hyperspectral imaging has emerged as a powerful remote sensing modality, extensively employed in defense [1], mineral mapping [2], vegetation health analysis [3], medical science [4], and oil spill detection [5]. HSI captures rich spectral information across hundreds of contiguous spectral bands, enabling detailed characterization and classification of materials at the pixel level. However, the high dimensionality and redundancy inherent in hyperspectral datasets pose significant challenges for accurate and efficient pixel-wise image classification [6]. Addressing these challenges requires sophisticated methods capable of effectively exploiting spectral information, reducing redundancy, and improving classification performance.

Traditional spectral feature extraction techniques, which often depend on heuristic or basic statistical criteria, frequently overlook nuanced spectral interactions, thus limiting their capability in classification tasks [7]. Recent breakthroughs in deep learning, particularly attention mechanisms initially introduced to capture dependencies in natural language processing [8], provide a foundation to effectively model complex spectral interrelations [9]. Inspired by the success of attention architectures in vision tasks [10], in this work, we propose an Attention-based Spectral Profile Representation (ASPR) network which employs attention modules to dynamically weight spectral bands, enabling precise identification and representation of discriminative spectral patterns. Unlike conventional approaches, ASPR facilitates richer spectral profile representation, ultimately enhancing classification accuracy and computational efficiency in hyperspectral image analysis.

The proposed ASPR method is specifically designed for pixel-wise hyperspectral image classification. Specifically, ASPR effectively leverages the attention scores of spectral bands to enable the model to focus on the most relevant spectral information. Additionally, recognizing the common issue of limited labeled hyperspectral data, we implement data augmentation techniques to bolster the training process, promoting model robustness and generalization. Experimental results demonstrate that our approach consistently outperforms existing methods, highlighting the efficacy and practical applicability of ASPR in real-world hyperspectral imaging scenarios.

In the rest of the paper, Section 2 summarizes related works. Section 3 introduces the proposed methodology, while the experimental results are presented in Section 4. Finally, we conclude the work in Section 5.

II. Related Works

Recent advances in hyperspectral image classification (HSIC) have predominantly leveraged deep learning techniques, capitalizing on their ability to handle complex, high-dimensional spectral data [11]. Early traditional approaches relied on dimensionality reduction strategies such as Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) to reduce computational complexity and spectral redundancy [12], [13]. However, these methods generally overlook the inherent nonlinear spectral-spatial characteristics of hyperspectral images.

With the introduction of Convolutional Neural Networks (CNNs), spatial-spectral feature extraction has significantly improved HSIC performance. Hu et al. [14] proposed a 1D-CNN architecture to extract spectral features directly, achieving enhanced classification performance. Similarly, Sidike et al. [15] introduced deep Progressive Expansion Networks

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA. Downloaded on April 24,2026 at 10:49:50 UTC from IEEE Xplore. Restrictions apply.

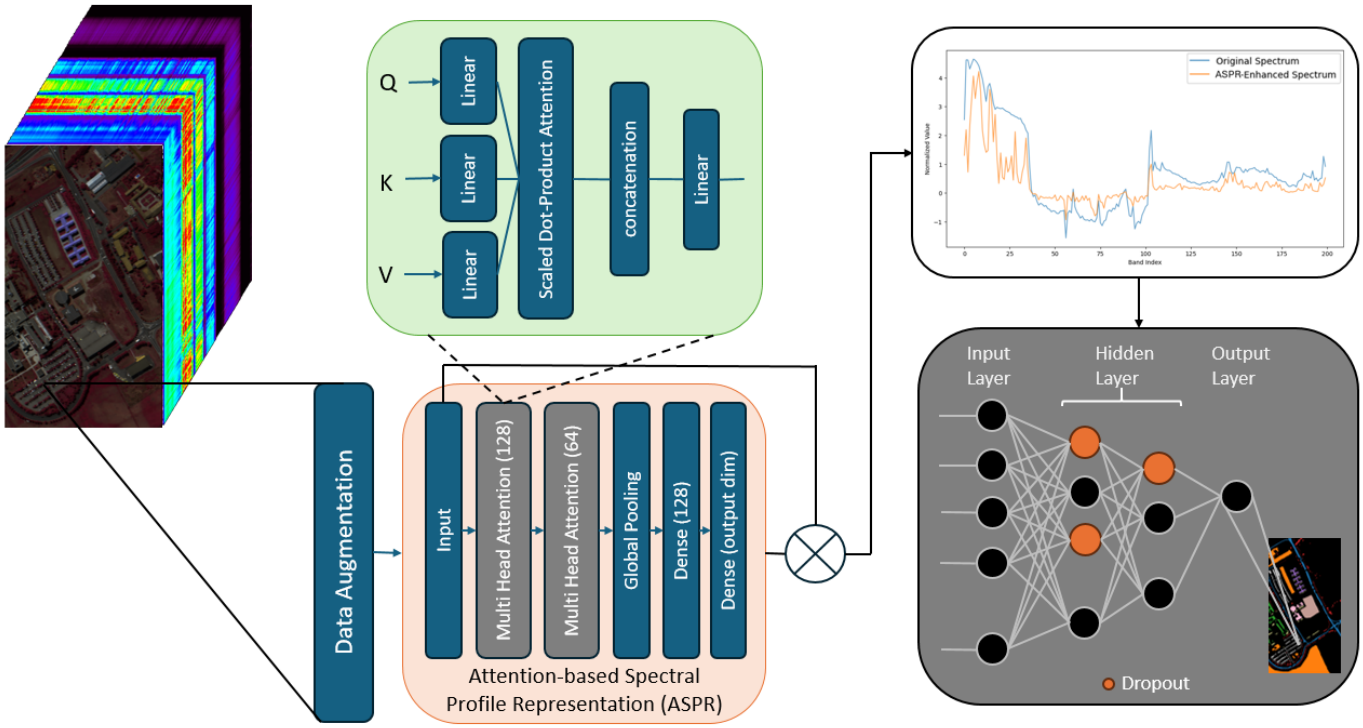

Fig. 1. Attention-based Spectral Profile Representation and Classifier Architecture

(dPEN), emphasizing progressive feature extraction through deeper CNN layers. These works demonstrate CNNs' capability to effectively capture spectral correlations, substantially outperforming traditional feature extraction methods.

Recently, attention mechanisms [10] have been extensively studied due to their ability to dynamically weigh spectral bands based on their relevance to the classification task. Attention-based CNN frameworks have proven to be effective in hyperspectral image classification by adaptively emphasizing discriminative spectral characteristics [16], [17]. Moreover, Hong et al. [9] showed better performance of the transformer model in capturing local spectral sequences from neighboring bands, compared to traditional CNN and standard transformers.

Data augmentation strategies have also been critical to addressing the limited labeled data challenge commonly observed in hyperspectral datasets. Techniques such as Gaussian noise injection, spatial augmentation, and spectral perturbation have been extensively used to enhance the generalization and robustness of DL models [18], [19]. These methods have consistently shown improvement in model performance, especially in scenarios with small training sets.

Building upon these developments, our proposed HSIC framework consists of ASPR, three fully-connected layers, data augmentation, and normalization techniques. The proposed method aims to overcome existing limitations by enhancing spectral discrimination and minimizing redundancy, thereby demonstrating improved classification accuracy and robustness over traditional and recent deep learning approaches.

III. Proposed Methodology

A. Proposed Architecture

1) Model Overview: Our core idea involves utilizing an attention-based spectral band representation block to evaluate and rank spectral bands based on their computed importance scores. Figure 1 shows the proposed workflow for HSIC, which begins with data augmentation, followed by an attention-based spectral profile representation (ASPR) using multi-head attention and pooling layers. The extracted attention scores are applied to the inputs to generate new features. The features are then classified through a neural network with dropout regularization to enhance generalization and improve classification performance.

Specifically, the ith spectral vector $\mathbf{x}_i \in \mathbb{R}^B$ is linearly projected into three vectors as follows:

$$Q = \mathbf{x}_i W^Q, \quad K = \mathbf{x}_i W^K, \quad V = \mathbf{x}_i W^V \tag{1}$$

where $W^Q, W^K, W^V \in \mathbb{R}^{B \times d_k}$ are learnable weight matrices corresponding to the query, key, and value projections. The term d_k refers to the dimension of the key vectors. Each of the h attention heads operates on its own set of Q, K, V matrices. The scaled dot-product attention for a head is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \tag{2}$$

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA. Downloaded on April 24,2026 at 10:49:50 UTC from IEEE Xplore. Restrictions apply.

The outputs from all heads are concatenated and projected back to the original dimension using an additional weight matrix. This enables the model to attend to different parts of the spectral signal simultaneously.

For instance, consider a hyperspectral image with $B = 200$ spectral bands and $N = 16{,}384$ pixels (e.g., 128×128), using $h = 4$ attention heads and $d_k = 32$. Each head projects the input into $Q, K, V \in \mathbb{R}^{16{,}384 \times 32}$, computes attention, and outputs a matrix of shape $16{,}384 \times 32$. The outputs from all heads are concatenated into a $16{,}384 \times 128$ matrix, which is then projected to $16{,}384 \times 200$.

After obtaining the attention outputs, we apply them to the original hyperspectral data to extract pertinent spectral features, expressed as

$$\mathbf{x}_i^a = \mathbf{x}_i \odot \mathbf{a}_i \tag{3}$$

where $\odot$ denotes the element-wise multiplication and a_i is the output of the ASPR block as shown in Figure 1.

$$\mathbf{X}_{\mathrm{a}} = \{\mathbf{x}_i^{\mathrm{a}}\}_{i=1}^{N} \tag{4}$$

where N denotes the number of samples in the dataset. As shown in Fig. 2, various samples after the implementation of ASPR demonstrate smoother and more stable spectral curves compared to the original spectra. The ASPR-based spectra effectively reduce noise and fluctuations, highlighting critical spectral features and enabling more reliable interpretation across most spectral bands.

2) Classification Layer: The classification layer employs a concise architecture with two sequential hidden dense layers: a first layer with 256 units and ReLU activation, followed by a second layer with 128 units (ReLU activation), and a final softmax layer for multiclass classification. The class imbalance problem is handled using the `compute_class_weight`function from scikit-learn. This function gives higher weights to less frequent classes, so the model pays more attention to them during training. This helps the model become more accurate and reliable. The model is compiled with the Adam optimizer and sparse categorical cross-entropy loss. This streamlined approach ensures efficient training and robust predictive performance.

B. Data Augmentation and Normalization

To further enhance model performance, the dataset undergoes a normalization procedure followed by data augmentation. Specifically, each spectral band was normalized by subtracting the band-specific mean and scaling by its standard deviation across all spectral vectors. Bands affected by water absorption (e.g., in Indian Pines and Salinas datasets) were removed prior to normalization. Additionally, Gaussian noise-based augmentation is applied to increase the variability of the training data.

C. Model Training

Training optimization is conducted using the Adam optimizer, and sparse categorical cross-entropy is utilized as the loss function. To ensure effective training, dynamic learning rate adjustment is implemented using the *ReduceLROnPlateau* callback, and optimal models were preserved via *ModelCheckpoint* callbacks.

D. Training and Evaluation

The model is trained on three popular hyperspectral datasets and evaluated by metrics such as overall accuracy (OA), average accuracy (AA), and Cohen's kappa coefficient.

E. Implementation and Software Tools

The proposed methodology is implemented using Python, primarily leveraging TensorFlow and Keras for deep learning models. Data preprocessing and evaluation tasks were performed with libraries such as NumPy, Scikit-learn, and SciPy. All experiments were conducted with reproducibility in mind, using fixed random seeds across TensorFlow and NumPy.

IV. Experimental Results

A. Datasets

The datasets used in our experiments are Indian Pines (IP), University of Pavia (UP) and Salinas Valley (SV), which are publicly available [1] and widely used for evaluation of HSI classification methods. The details of the datasets are described as follows.

Indian Pines (IP): The Indian Pines dataset was collected using an AVIRIS sensor over the Indian Pines area located in Northwestern Indiana. The acquired imagery includes 145×145 pixels with a spectral resolution comprising 224 bands spanning wavelengths from 400 nm to 2500 nm. In this dataset, agricultural areas constitute roughly two-thirds, while forests and natural perennial vegetation make up the remaining one-third. Ground-truth annotations categorize the data into 16 different classes, though these classes are not entirely distinct and may have overlapping characteristics. After removing spectral bands affected by water absorption, 200 usable bands are retained. Figures 3(a) and (b) illustrate a pseudocolor representation of the image and the corresponding ground-truth classification map for the Indian Pines dataset, respectively.

University of Pavia (UP): As part of the Pavia scenes, the University of Pavia HSI dataset was acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over the city of Pavia, located in northern Italy. The size of the image is 610×340 with a total of 103 bands. There are nine types of land cover with a spatial resolution of 1.3 meters. Figures 4 (a) and (b) show the pseudocolor image and the ground truth map of the UP dataset, respectively.

Salinas Valley (SV): The Salinas Valley HSI dataset was collected by the AVIRIS sensor over the Salinas Valley in California. The dataset comprises 512×217 pixels with a spatial resolution of 3.7-meter pixels, and contains a total of 220 spectral reflectance bands. In this dataset, the water absorption bands (20 in total) were also discarded, which left

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA. Downloaded on April 24,2026 at 10:49:50 UTC from IEEE Xplore. Restrictions apply.

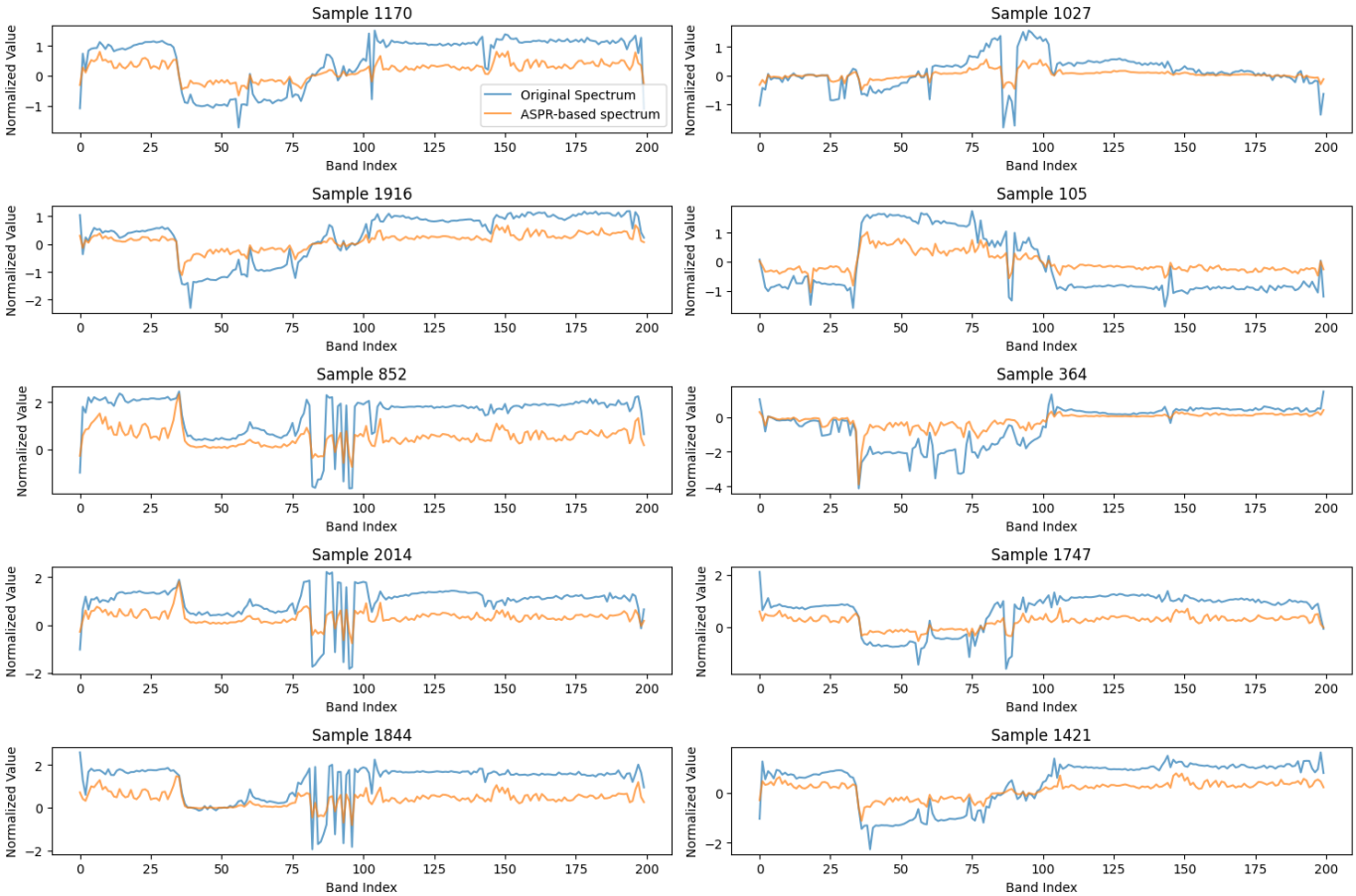

Fig. 2. Sample of Original and ASPR from Indian Pines Dataset.

TABLE I
ACCURACY (IN %) COMPARISON WITH DIFFERENCES FROM ASPR ON THE INDIAN PINES DATASET, VALUES IN PARENTHESES (+/–) INDICATE THE DIFFERENCE FROM ASPR

Metric	**1D-CNN** [14]	**dPEN** [15]	**Plastic-Net** [20]	**GAF U-Net** [21]	**GAF-NAU** [21]	**ASPR**
OA (%)	74.39(-7.27)	77.62(-4.04)	73.00(-8.66)	75.67(-5.99)	81.07(-0.59)	**81.66**
AA (%)	76.36(-2.43)	**81.28(+2.49)**	63.29(-15.50)	71.69(-7.10)	74.67(-4.12)	78.79
κ	70.57(-8.53)	74.54(-4.56)	69.23(-9.87)	71.89(-7.21)	78.31(-0.79)	**79.10**

TABLE II
ACCURACY (IN %) COMPARISON WITH DIFFERENCES FROM ASPR ON THE UNIVERSITY OF PAVIA DATASET. VALUES IN PARENTHESES (+/–) INDICATE THE DIFFERENCE FROM ASPR

Metric	**1D-CNN** [14]	**dPEN** [15]	**Plastic-Net** [20]	**GAF U-Net** [21]	**GAF-NAU** [21]	**ASPR**
OA (%)	90.17(-4.72)	91.21(-3.68)	89.70(-5.19)	89.70(-5.19)	91.12(-3.77)	**94.89**
AA (%)	89.37(-4.53)	90.54(-3.36)	87.76(-6.14)	87.70(-6.20)	90.49(-3.41)	**93.90**
κ	86.85(-6.39)	88.29(-4.95)	86.29(-6.95)	86.25(-6.99)	88.09(-5.15)	**93.24**

for 204 bands. The provided ground truth contains a total of 16 classes that cover regions of vegetables, bare soils, and vineyard fields. Figure 5 (a) and (b) shows the pseudocolor image and the corresponding ground truth map, respectively.

B. Results

The performance of our proposed model is comprehensively evaluated against several state-of-the-art deep learning methods for pixel-wise HSIC, including 1D-CNN [14], dPEN [15], Plastic-Net [20], GAF U-Net [21], and GAF-NAU [21]. As

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA. Downloaded on April 24,2026 at 10:49:50 UTC from IEEE Xplore. Restrictions apply.

TABLE III
ACCURACY (IN %) COMPARISON WITH DIFFERENCES FROM ASPR ON THE SALINAS DATASET. VALUES IN PARENTHESES (+/–) INDICATE THE DIFFERENCE FROM ASPR

Metric	**1D-CNN** [14]	**dPEN** [15]	**Plastic-Net** [20]	**GAF U-Net** [21]	**GAF-NAU** [21]	**ASPR**
OA (%)	91.43(-1.79)	92.53(-0.69)	90.69(-2.53)	93.82(+0.60)	**94.59(+1.37)**	93.22
AA (%)	95.06(-1.42)	96.48(+0.00)	95.06(-1.42)	96.57(+0.09)	**97.00(+0.52)**	96.48
κ	90.45(-1.99)	91.68(-0.76)	89.63(-2.81)	93.11(+0.67)	**93.97(+1.53)**	92.44

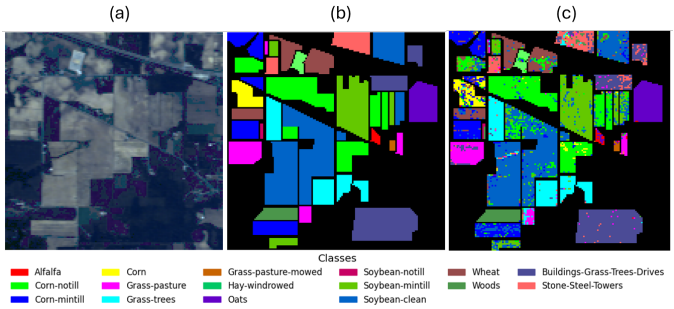

Fig. 3. Indian Pines hyperspectral dataset with classification maps. (a) True color composite of HSI, (b) Ground truth. (c) ASPR-based classification.

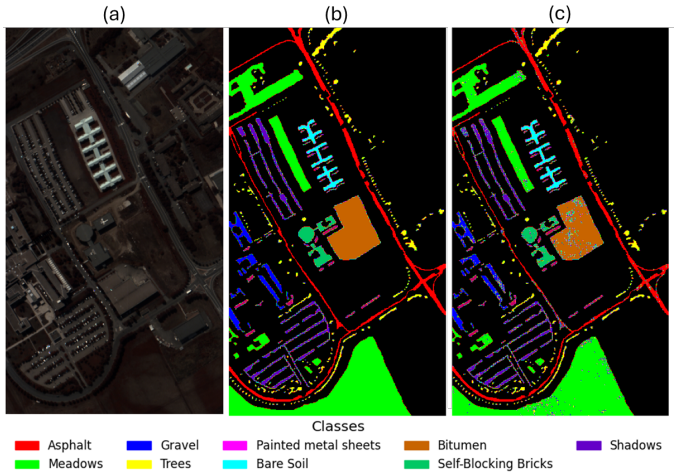

Fig. 4. University of Pavia scene hyperspectral dataset with classification maps. (a) True color composite of HSI, (b) Ground truth. (c) ASPR-based classification.

illustrated in Tables I, III, and II, the ASPR-based method demonstrated superior performance across the aforementioned evaluation metrics on the IP and UP datasets, while exhibiting slightly lower accuracy on the SV dataset. This highlights its robustness in handling complex spectral characteristics. The observed values of OA, AA, and Cohen's Kappa coefficient further underscore ASPR's effectiveness in accurately distinguishing subtle spectral differences compared to other benchmark methods.

The classification maps of IP, UP, and SV datasets are presented in Figures 3, 4, 5, respectively. For each dataset, subfigure (a) shows the true-color composite of the original

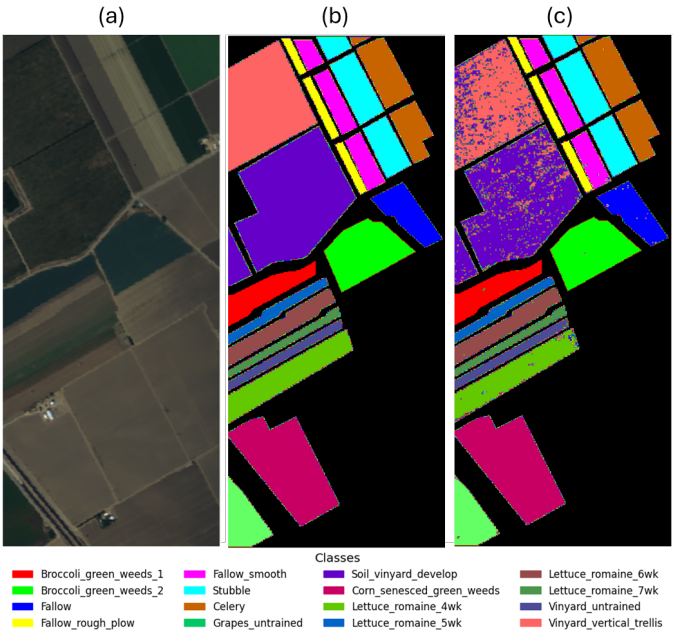

Fig. 5. Salinas Valley hyperspectral dataset with classification maps. (a) True color composite of HSI, (b) Ground truth. (c) ASPR-based classification.

hyperspectral image, (b) presents the ground truth labels, and (c) displays the classification output of the ASPR method. ASPR's results generally align with the ground truth, though some misclassifications are noticeable, particularly in heterogeneous regions that are challenging to classify, while still enhancing spatial smoothness.

V. CONCLUSION

In this study, we proposed the Attention-Based Spectral Profile Representation (ASPR) framework for hyperspectral image classification, utilizing multi-head attention mechanisms, data augmentation, and simple fully-connected layers to enhance performance. The proposed ASPR technique effectively identifies and preserves essential spectral information while significantly reducing redundancy, thereby enhancing classification accuracy and computational efficiency. Comprehensive evaluations against state-of-the-art methods demonstrated ASPR's superior or highly competitive performance. Our results affirm ASPR's effectiveness in capturing intricate spectral correlations, generalizing well across various hyperspectral imaging scenarios, and effectively addressing the inherent challenges of hyperspectral image classification.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA. Downloaded on April 24,2026 at 10:49:50 UTC from IEEE Xplore. Restrictions apply.

REFERENCES

[1] P. W. Yuen and M. Richardson, "An introduction to hyperspectral imaging and its application for security, surveillance and target acquisition," *The Imaging Science Journal*, vol. 58, no. 5, pp. 241–253, 2010.

[2] N. Yokoya, J. C.-W. Chan, and K. Segl, "Potential of resolution-enhanced hyperspectral data for mineral mapping using simulated enmap and sentinel-2 images," *Remote Sensing*, vol. 8, no. 3, p. 172, 2016.

[3] L. Liang, L. Di, L. Zhang, M. Deng, Z. Qin, S. Zhao, and H. Lin, "Estimation of crop lai using hyperspectral vegetation indices and a hybrid inversion method," *Remote Sensing of Environment*, vol. 165, pp. 123–134, 2015.

[4] U. Khan, S. Paheding, C. P. Elkin, and V. K. Devabhaktuni, "Trends in deep learning for medical hyperspectral image analysis," *IEEE Access*, vol. 9, pp. 79 534–79 548, 2021.

[5] M. S. Alam and P. Sidike, "Trends in oil spill detection via hyperspectral imaging," in *2012 7th International Conference on Electrical and Computer Engineering*. IEEE, 2012, pp. 858–862.

[6] G. Camps-Valls, M. Campos-Taberner, Álvaro Moreno-Martínez, S. Walther, G. Duveiller, A. Cescatti, M. D. Mahecha, J. Muñoz-Marí, F. J. García-Haro, L. Guanter, M. Jung, J. A. Gamon, M. Reichstein, and S. W. Running, "A unified vegetation index for quantifying the terrestrial biosphere," *Science Advances*, vol. 7, no. 9, p. eabc7447, 2021. [Online]. Available: https://www.science.org/doi/abs/10.1126/sciadv.abc7447

[7] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geoscience and Remote Sensing Magazine*, vol. 7, no. 2, pp. 118–139, 2019.

[8] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[9] D. Hong, Z. Han, J. Yao, L. Gao, B. Zhang, A. Plaza, and J. Chanussot, "Spectralformer: Rethinking hyperspectral image classification with transformers," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–15, 2022.

[10] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv:2010.11929*, 2020.

[11] S. K. Chandrappa and S. Paheding, "Exploring the depth of the kan method for hyperspectral image classification," in *Proceedings of the 2025 ASEE Northeast Section Conference*. University of Bridgeport, Bridgeport, CT: ASEE Conferences, mar 2025, https://peer.asee.org/55021.

[12] M. Fauvel, J. Chanussot, and J. A. Benediktsson, "Advances in spectral-spatial classification of hyperspectral images," *Proceedings of the IEEE*, vol. 101, no. 3, pp. 652–675, 2013.

[13] L. Zhang, Q. Liu, and D. Zhang, "A review of hyperspectral image classification and feature extraction," *IEEE Geoscience and Remote Sensing Magazine*, vol. 4, no. 2, pp. 31–46, 2016.

[14] W. Hu, Y. Huang, W. Li, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *Journal of Sensors*, vol. 2015, pp. 1–12, 07 2015.

[15] P. Sidike, V. Sagan, M. Maimaitijiang, M. Maimaitiyiming, N. Shakoor, J. Burken, T. Mockler, and F. B. Fritschi, "dpen: deep progressively expanded network for mapping heterogeneous agricultural landscape using worldview-3 satellite imagery," *Remote Sensing of Environment*, vol. 221, pp. 756–772, 2019. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0034425718305406

[16] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel resnet for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 9, pp. 7831–7843, 2020.

[17] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided cnns," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 3, pp. 2281–2293, 2020.

[18] H. Yu and Q. Du, "Data augmentation for hyperspectral image classification with deep cnn," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 12, pp. 1795–1799, 2017.

[19] J. Nalepa, M. Myller, and M. Kawulok, "Data augmentation for hyperspectral image classification with convolutional neural networks," *IEEE Geoscience and Remote Sensing Letters*, vol. 17, no. 3, pp. 491–495, 2020.

[20] S. Jiang, Z. Xu, M. Kamran, S. Zinchik, S. Paheding, A. G. McDonald, E. Bar-Ziv, and V. M. Zavala, "Using atr-ftir spectra and convolutional neural networks for characterizing mixed plastic waste," *Computers & Chemical Engineering*, vol. 155, p. 107547, 2021.

[21] S. Paheding, A. A. Reyes, A. Kasaragod, and T. Oommen, "Gafnau: Gramian angular field encoded neighborhood attention u-net for pixel-wise hyperspectral image classification," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, June 2022, pp. 409–417.

Authorized licensed use limited to: NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA. Downloaded on April 24,2026 at 10:49:50 UTC from IEEE Xplore. Restrictions apply.